

Introduction

Fitz Koch

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Overview

This mini-website will serve as an introduction to the skills and projects we'll be working on during the summer of 2026 for the Okemo Verso Pod.

Skills

1. `Git` is really great for working on projects together as well as for storing your stuff cheaply. Hopefully VERSO's main onboarding helped with that.
2. Quarto Notebooks (`.qmd`) give you the advantages of jupyter notebooks in a more replicable and git-friendly way:
 - Incredibly versatile, and can include any `Python` code as well as `R`.
 - Interactive coding environment
 - One document sources converts to:
 - Interactive slides
 - `.html` for websites, just like this one
 - `.ipynb` jupyter notebook files
 - `.pdf` files. Note this can use Latex, like: $2 = \frac{10}{5}$
 - Are purely text-based, so git differentials make sense and are actually readable.
3. Coding practices
 - Using a project-wide formatter. We'll use `ruff`.
 - Keeps code readable and short
 - Minimizes unnecessary `git` differentials.
 - Proper `git` usage
 - commit on every meaningful change
 - pull request every complete feature.
 - respond to, and fix, reviews.

- Type hints
 - clearer code
 - easier bug hunts
4. Data Engineering for larger-scale data-science
 - ETL
 - See how data works when you have a lot of it that you want to use at different times, many times, in different ways.
 5. Policy Research
 - Building good projects often means translating prior work or uncertain requests. This will help with that.
 6. Typescript (Optional)
 - learn true power of types
 - learn how internet actually works

Projects

1. Benefits Cliff
 - Languages: `qmd` files (no code) and `typescript` logic
 - Goal: help legislators and everyday people understand:
 - what benefits are available
 - how those benefits change with income or family situation
 - how those benefits relate with census information about income and family situation
 - Work to be done:
 - Adding more benefits
 - * Policy review: build out core logic in `qmd` files for calculating benefits from policy manuals and other resources
 - * typescript: code up algorithms
 - * backfill: work through existing typescript algorithms and backfill the logic into `qmd` documentation.
 - Part 2: Extension
 - * Add in information about resources required for life per county
 - * Data analysis: pull in census information, etc., and see how comparisons work out.
2. Vermont Reports
 - Languages: `SQL` (`duckdb`), `qmd`, and `Python`.

- Goal: help citizens, select boards, local government, and planners better understand their towns
 - Work to be done:
 - Data Engineering:
 - * Updating all data engineering logic :
 1. get data (external APIs or raw download)
 2. build clean data (SQL and python scripts)
 - requires lots of thought and consultation about how we should most effectively split up the tables.
 3. runtime queries / logic (SQL and python scripts)
 4. api (pythons' FASTAPI library)
 5. frontend
 - * Removing and cleaning legacy code
 - Analysis:
 - * Adding in new functionality and places for analysis, especially putting different data streams in conversation with each other
 - EG, how broadband and wastewater infrastructure inter-relate.
3. TBD:
- Languages: almost certainly **SQL**, **Python**, **.qmd**
 - Goal: help people of Vermont. Working with Kendall to see exactly what that is.

Next Steps

Complete the following tutorials:

1. Git: complete standard VERSO repository.
2. Complete the [Data Engineering In Quarto](#) project
3. Complete the [Basic Typescript](#) project (COMING SOON)